Phishing Awareness Analysis

DecodeLabs Cyber Security Internship – Project 3

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Cybersecurity

Abstract

Phishing is one of the most common cyberattacks used by criminals to steal confidential information from individuals and organizations. Attackers often disguise themselves as trusted entities and send fraudulent emails or messages to deceive users. The objective of this project is to identify phishing attempts by analyzing suspicious emails and comparing them with legitimate communications.

This project focuses on recognizing phishing indicators, identifying suspicious links, understanding social engineering techniques, and increasing cybersecurity awareness.

Introduction

Cybercriminals use various methods to gain unauthorized access to sensitive information. Among these methods, phishing remains one of the most dangerous and widely used attacks. These attacks rely on human error rather than technical vulnerabilities.

Phishing attacks usually include:

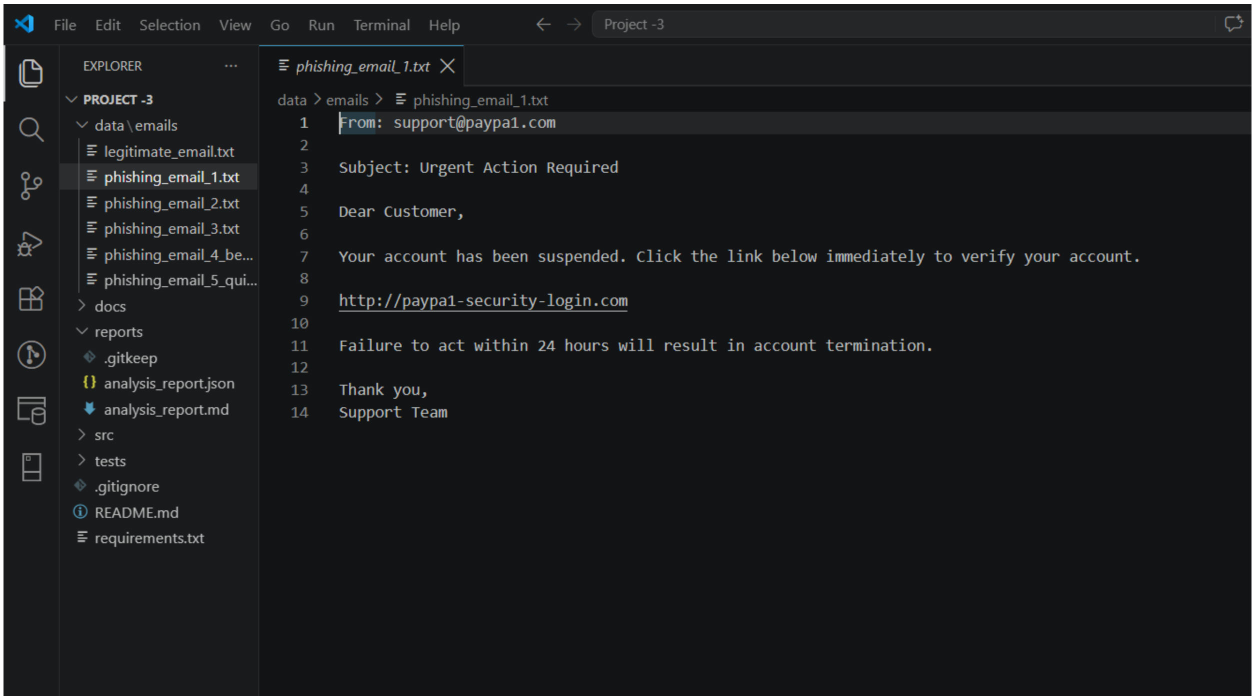
- Fake sender addresses
- Suspicious hyperlinks
- Requests for sensitive information
- Threatening or urgent language
- Spelling and grammatical errors

Understanding these indicators is essential for maintaining information security.

Project Objectives

The objectives of this project are as follows:

- Analyze suspicious emails.
- Identify phishing indicators.
- Detect malicious links.
- Distinguish legitimate emails from fraudulent messages.
- Improve cybersecurity awareness.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Project -3
EXPLORER
PROJECT -3
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legitimate_email.txt
phishing_email_1.txt
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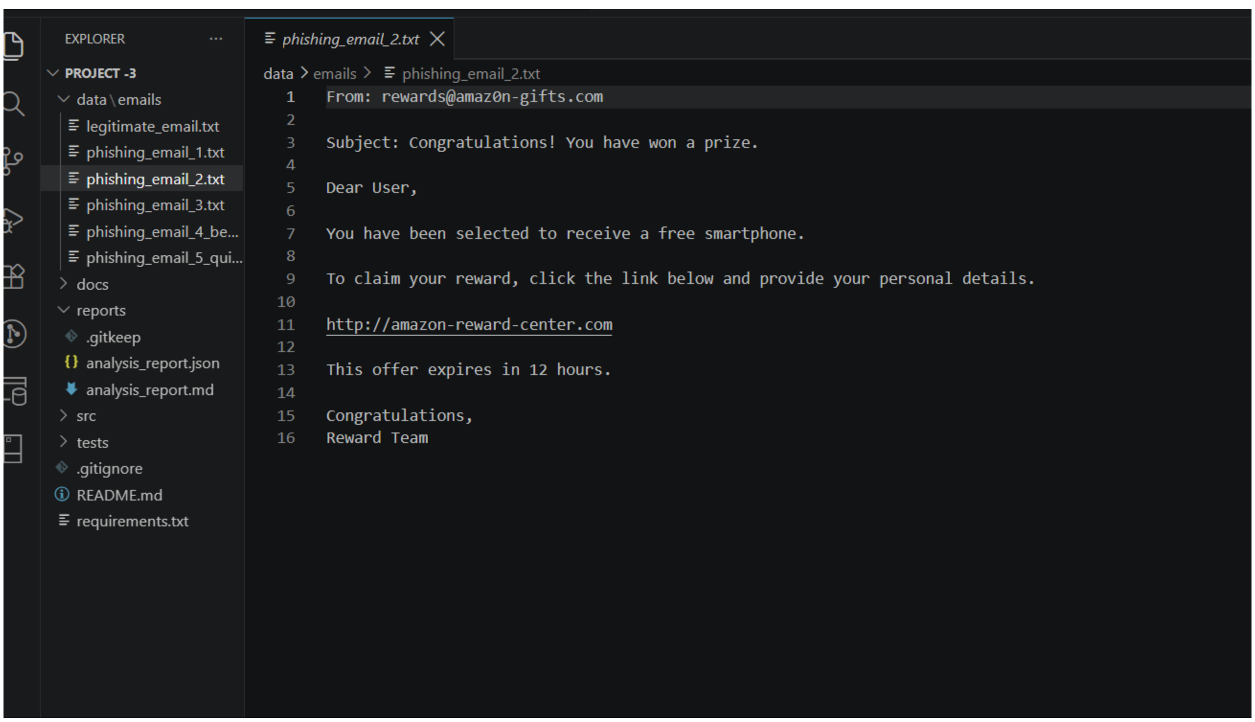
reports
.gitkeep
analysis_report.json
analysis_report.md
src
tests
.gitignore
README.md
requirements.txt
phishing_email_1.txt
data > emails > phishing_email_1.txt
From: support@paypa1.com
Subject: Urgent Action Required
Dear Customer,
Your account has been suspended. Click the link below immediately to verify your account.
http://paypa1-security-login.com
Failure to act within 24 hours will result in account termination.
Thank you,
Support Team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PROJECT -3
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legitimate_email.txt
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reports
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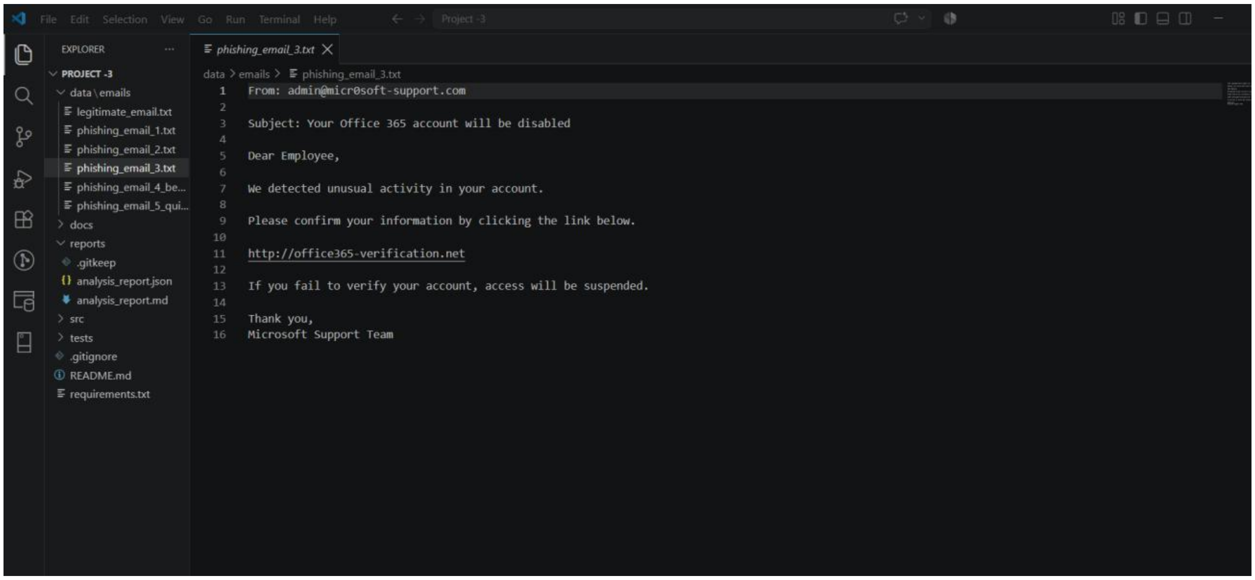
analysis_report.json
analysis_report.md
src
tests
.gitignore
README.md
requirements.txt
phishing_email_2.txt
data > emails > phishing_email_2.txt
From: rewards@amaz0n-gifts.com
Subject: Congratulations! You have won a prize.
Dear User,
You have been selected to receive a free smartphone.
To claim your reward, click the link below and provide your personal details.
http://amazon-reward-center.com
This offer expires in 12 hours.
Congratulations,
Reward Team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data \ emails
legitimate_email.txt
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phishing_email_2.txt
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phishing_email_4_be...
phishing_email_5_qui...
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reports
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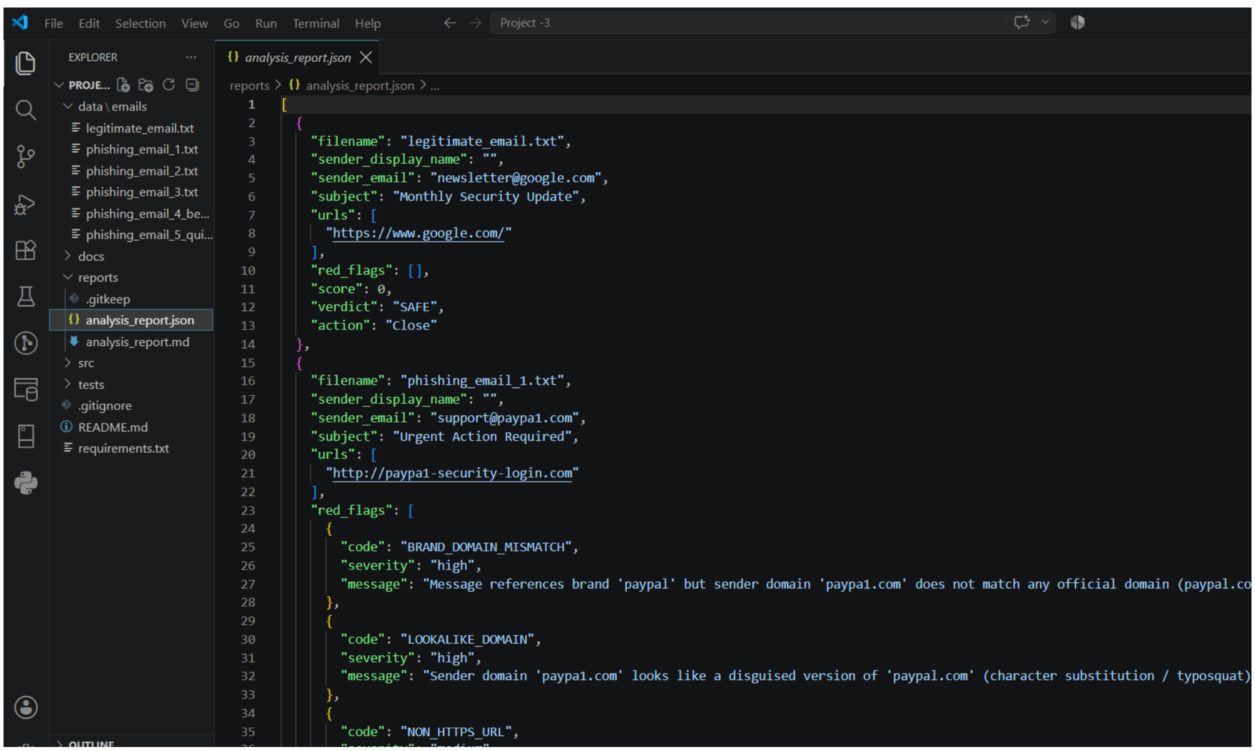
analysis_report.json
analysis_report.md
src
tests
.gitignore
README.md
requirements.txt
phishing_email_3.txt
data > emails > phishing_email_3.txt
From: admin@micr0soft-support.com
Subject: Your Office 365 account will be disabled
Dear Employee,
We detected unusual activity in your account.
Please confirm your information by clicking the link below.
http://office365-verification.net
If you fail to verify your account, access will be suspended.
Thank you,
Microsoft Support Team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EXPLORER
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data \ emails
legitimate_email.txt
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phishing_email_4_be...
phishing_email_5_qui...
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reports
.gitkeep
analysis_report.json
analysis_report.md
src
tests
.gitignore
README.md
requirements.txt
OUTLINE
analysis_report.json
reports > analysis_report.json > ...
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"sender_display_name": "",
"sender_email": "newsletter@google.com",
"subject": "Monthly Security Update",
"urls": [
"https://www.google.com/"
],
"red_flags": [],
"score": 0,
"verdict": "SAFE",
"action": "Close"
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"filename": "phishing_email_1.txt",
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"urls": [
"http://paypa1-security-login.com"
],
"red_flags": [
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"message": "Message references brand 'paypal' but sender domain 'paypa1.com' does not match any official domain (paypal.co
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"code": "LOOKALIKE_DOMAIN",
"severity": "high",
"message": "Sender domain 'paypa1.com' looks like a disguised version of 'paypal.com' (character substitution / typosquat)
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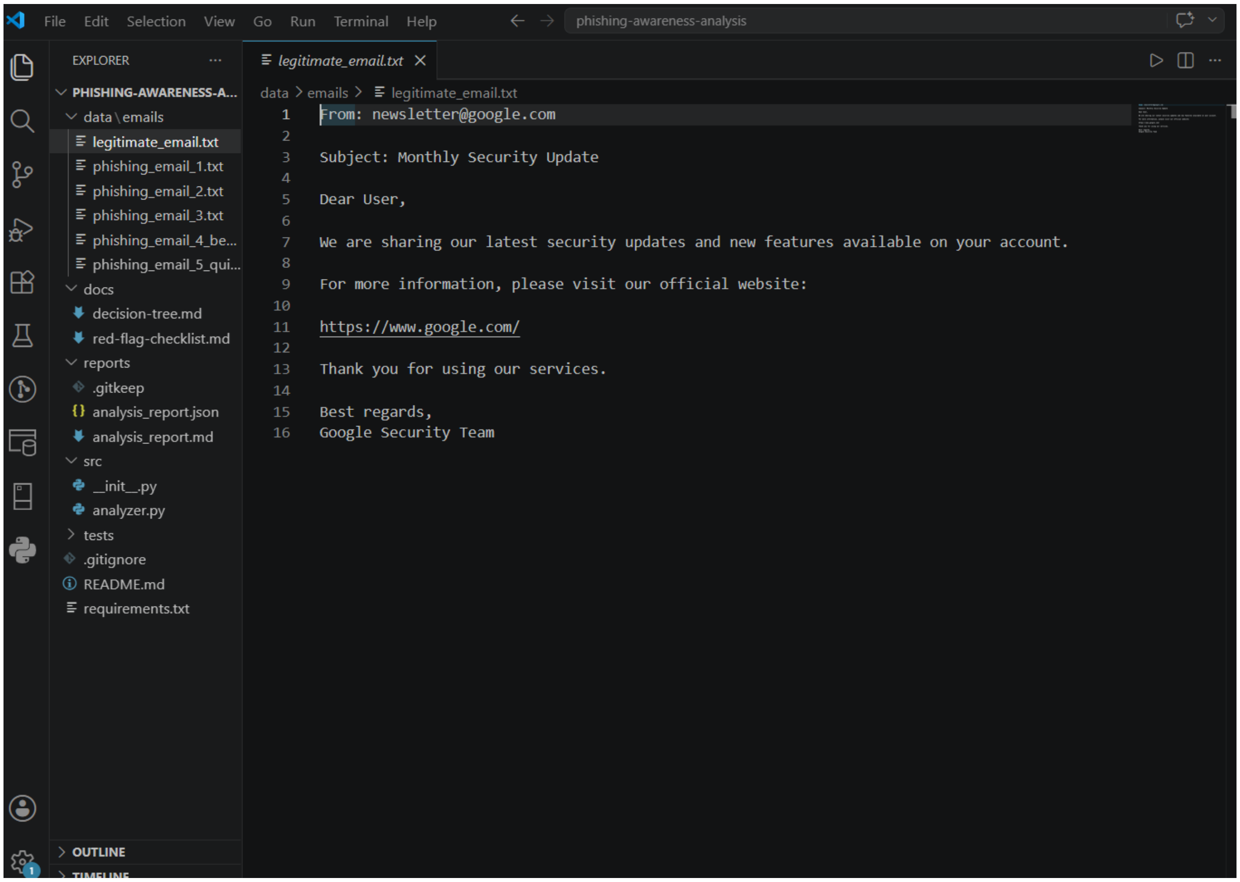
"code": "NON_HTTPS_URL",



Tools Used

- Microsoft Word
- Notepad
- Virus Total
- URLScan.io
- Snipping Tool
- GitHub

Skills Acquired

- Threat analysis
- Risk assessment
- Cybersecurity awareness
- Social engineering analysis
- Email security analysis

Results

The project successfully identified multiple phishing attempts and highlighted the techniques used by attackers to deceive users. By comparing legitimate and fraudulent emails, it became easier to understand how cybercriminals manipulate human behavior.

Conclusion

This project provided practical experience in recognizing phishing attacks and understanding their potential impact. The knowledge gained from this project can help users protect themselves from malicious activities and strengthen their overall cybersecurity awareness.

Continuous learning and careful analysis are essential for identifying and preventing future attacks.